

Derivada Covariante Antissimetrizada

$$V = V_i e^i$$

$$V_{[i;jk]} = \frac{1}{2} [V_{ijk} - V_{kji}]$$

$$= \frac{1}{2} \left(V_{ijk} - \Gamma^m_{ik} V_m - V_{k,i} + \Gamma^m_{ki} V_m \right)$$

$$= \frac{1}{2} (V_{ijk} - V_{k,i}) + \frac{1}{2} (\Gamma^m_{ik} - \Gamma^m_{ki})$$

= 0 se conexão for o símbolo de christoffel

é um tensor

Logo:

$$V_{ijk} - V_{kji} = V_{ijk} - V_{k,i} \text{ se } \Gamma \text{ for simétrica}$$

Tensor de Riemann

$$V^i_{j[k;m]} = V^i_{jkm} \quad \text{e} \quad 2V^i_{j[km;n]} = V^i_{;km} - V^i_{jmn}$$

$$2V^i_{j[km;n]} = (\Gamma^i_{nkm} - \Gamma^i_{mn,k} + \Gamma^i_{em} \Gamma^e_{nk} - \Gamma^i_{ek} \Gamma^e_{nm}) V^n + \underbrace{(\Gamma^m_{mn} - \Gamma^m_{km})}_{2\Gamma^m_{[mn]} = 0} V^i_{jn}$$

Define-se:

$$R^i_{nmk} = (\Gamma^i_{nkm} - \Gamma^i_{mn,k} + \Gamma^i_{em} \Gamma^e_{nk} - \Gamma^i_{ek} \Gamma^e_{nm})$$